

MANI SHARATH CHANDRA SATTU

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Summary

Data Science graduate with 2+ years of software engineering experience in data analysis, pipeline development, and machine learning using Python, SQL, and R. Skilled in building scalable workflows, creating dashboards in Tableau and Power BI, and delivering actionable insights. Seeking data analytics or engineering roles to drive business impact through data-driven solutions.

Work Experience

Grand Valley State University, College of Computing

Jan 2025-Apr 2025

Research Assistant

Allendale, USA

- Engineered deep learning pipelines using Python, PyTorch, and Vision Transformers (ViT) to classify COVID-19 from medical images, optimizing data quality through advanced preprocessing (resizing, normalization, augmentation).
- Enhanced model transparency by implementing Explainable AI (XAI) techniques (Grad-CAM, LIME, SHAP) and achieved 99.27% test accuracy through rigorous evaluation metrics.
- Led literature review and research synthesis on XAI methods using Zotero, collaborated cross-functionally to refine models, and co-authored academic deliverables advancing explainable healthcare AI solutions.

Blackstraw.ai

May 2021-July 2023

Software Engineer

Chennai, India

- Enhanced data workflows by refining Python and SQL code, improving data aggregation efficiency and reducing operational bugs.
- Built and optimized data parsers to address quality gaps, boosting data extraction accuracy and system performance.
- Led end-to-end data engineering processes, mentored new team members on analysis, QA, and coding standards, and delivered actionable insights to support strategic decision-making.

Projects

LightGlue- Local Feature Matching at Light Speed

Jan-2025

Reviewed and analyzed the LightGlue research paper to understand the architecture, methodology, and evaluation techniques. Contributed to project documentation and presentation by summarizing model innovations, results, and comparisons with baseline models like SuperGlue. Interpreted performance metrics such as precision, recall, and AUC across multiple benchmark datasets. Proposed potential extensions like temporal consistency and self-supervised learning to improve model generalization.

IPL Data Analysis and Visualization

Aug-2024

Collaborated on a project focused on visualizing and analyzing IPL cricket statistics to support team analysts, coaches, and auction advisors in strategic decision-making. Developed interactive dashboards using Python and Tableau to highlight top-performing batsmen and bowlers economy rates, compare team performances across different innings, and provide insights to facilitate player selection and team strategies for upcoming auctions.

Linear Regression for House Price Prediction

Jan-2024

Built a statistical model using linear regression to predict house prices with 90% accuracy, integrating data into a data management system to provide valuable insights into real estate market trends. Conducted thorough data analysis, preprocessing, and feature selection using R, with a focus on ensuring data quality through proper systems management.

Skills

- Programming Languages:** Python, R, Java
- Data Science:** Data Analysis, Data Engineering, Data Visualization, Explainable AI Methods
- Tools:** PowerBI, MS-Excel, Tableau, Git, Canva
- Database:** MySQL, PostgreSQL, MongoDB

Education

Anubose Institute of Technology

Jun 2017 - Sep 2020

Bachelor of Technology, Computer Science

Grand Valley State University

Aug 2023 - Apr 2025

Master of Science, Data Science and Analytics